# B. Perfect Number

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: binary search, brute force, dp, implementation, number theory |
https://codeforces.com/problemset/problem/919/B

We consider a positive integer perfect, if and only if the sum of its digits is exactly 10. Given a positive integer *k*, your task is to find the *k*-th smallest perfect positive integer.

## Input

A single line with a positive integer *k* ($1 \le k \le 10\,000$).

## Output

A single number, denoting the *k*-th smallest perfect integer.

## Examples

| Input |
| --- |
| 1 |

| Output |
| --- |
| 19 |

| Input |
| --- |
| 2 |

| Output |
| --- |
| 28 |

## Note

The first perfect integer is 19 and the second one is 28.